Certificate

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Institution GDC Doda

This is certified to be the bonafide work of the student in the Computer Application *Laboratory during the academic year* 2025/2026

No. of practicals certified 18 *out of* 21 *in the subject of* OBMS

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Teacher In-charge

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(N.B: The candidate is expected to retain his/her journal till he/she passes in the subject.)

Expt. No. 1 Date

Page No. 1

Aim: To study and implement DDL commands in SQL for creating tables.

Theory: DDL (Data Definition Language) commands are used to define and manage the structure of database objects such as tables.
The main DDL commands are:

- Create, Alter, Drop, Truncate

Description:

- Int is used for integer values.
- VARCHAR is used for storing text.
- CREATE TABLE creates a new table named Student.

Scol

```
CREATE TABLE Student (
  RollNo  INT,
  Name    VARCHAR (50),
  Class   VARCHAR (20)
  Marks   INT
);
```

output:

table Student created Successfully.

Expt. No. 2 Date

Page No. 3

Aim: To study SQL tools and different data types used in SQL.

Theory:

SQL (Structured Query language) is used to create and manage database.

SQL tools such as MySQL are used to execute SQL commands

Different data types in SQL are used to store different kinds of data.

Common SQL Data types:

Data type	Description
• INT	Stores integer values
• VARCHAR	Stores text values
• CHAR	Stores fixed-length characters
• DATE	Stores date values
• FLOAT	Stores decimal numbers.

Teacher's Signature

SQL

```
CREATE TABLE Student (
    RollNo INT,
    Name VARCHAR(50),
    Age INT,
    DOB DATE,
    Percentage FLOAT
);
```

output:

Table Student created successfully

Aim:
To create a table schema and insert records into the table using SQL commands.

Theory:
A Schema defines the structure of a table. The INSERT INTO command is used to add records into a table.

Description:
- CREATE TABLE creates the Employee table.
- INSERT INTO adds records into the table.

Teacher's Signature

SQL

```
CREATE TABLE Emoployee (
    Emp ID INT,
    Name VARCHAR(50),
    Department Varchar(30),
    Solary INT
);
INSERT INTO EmPloyee VALUES
(1, 'Ali', 'HR', 25000);
(2, 'Ahmed', 'IT', 40000);
(3, 'Zaid', 'Finance', 35000);
(4, 'Umar', 'Marketing', 30000);
(5, 'Bilal', 'Sales', 28000);
```

OutPut

EmpID	Name	Department	Solary
1	Ali	HR	25000
2	Ahmed	IT	40000
3	Zaid	Finance	35000
4	Umar	Marketing	30000
5	Bilal	Sales	28000

Date

Expt. No. 4 Page No. 4

Aim:
To study ALTER TABLE and DROP TABLE commands in SQL.

Theory:
The ALTER TABLE command is used to modify the structure of an existing table.
The ADD clause adds a new column, while the MODIFY clause changes the datatype or size of a column.
The DROP TABLE command deletes a table permanently.

Description:
- ADD adds a new column named Address.
- MODIFY changes the size of the Name column.
- DROP TABLE deletes the Student table.

Teacher's Signature

SQL

```
CREATE TABLE Student (
RollNo INT,
Name VARCHAR(30),
Marks INT
);

ALTER TABLE Student
ADD Address VARCHAR(50);

ALTER TABLE Student
MODIFY Name VARCHAR(50);

DROP TABLE Student;
```

Date

Expt. No. 5 Page No. 5

Aim:
To Study DROP, RENAME, TRUNCATE, and DESCRIBE commands in SQL.

Theory:
These SQL commands are used for managing tables in a database.

- DROP deletes a table Permanently.
- RENAME changes the table name.
- TRUNCATE removes all records from a table.
- DESCRIBE displays the structure of a table.

Teacher's Signature

SQL

```
CREATE TABLE Employee (
    EmpID INT,
    Name VARCHAR(50),
    Salary INT
);

DESCRIBE Employee;

RENAME TABLE Employee TO Staff;

TRUNCATE TABLE Staff;

DROP TABLE Staff;
```

Date

Expt. No. 6 Page No. 6

Aim:
To Study different SQL constraints used in database tables.

Theory:
Constraints are rules applied to table columns to maintain accuracy and integrity of data.

Common SQL Constraints:

- NOT NULL → Prevents null values
- UNIQUE → allows only unique values
- Primary Key → uniquely identifies each records.
- FOREIGN KEY → links two tables
- CHECK → limits values in a column
- Default → sets a default value.

Teacher's Signature

```
CREATE TABLE Department (
    DeptID INT PRIMARY KEY,
    DeptName VARCHAR(30) UNIQUE
);
CREATE TABLE Employee (
    EmpID INT PRIMARY KEY,
    Name VARCHAR(50) NOT NULL,
    Age INT CHECK (Age >= 18),
    Salary INT DEFAULT 20000,
    DeptID INT,
    FOREIGN KEY (DeptID)
    REFERENCES Department(DeptID)
);
```

Expt. No. 7 Date

Page No. 7

Aim:

To Study DML Commands in SQL.

Theory:

DML (Data Manipulation Language) commands are used to manage data in database tables.

Main commands in DML:

- INSERT → adds records
- SELECT → displays records
- UPDATE → modifies records
- DELETE → removes records.

Teacher's Signature

```
CREATE TABLE Student (
    RollNo INT,
    Name VARCHAR(50),
    Marks INT
);
INSERT INTO Student VALUES
(1, 'Ali', 85),
(2, 'Ahmed', 90),
(3, 'Zaid', 78);

SELECT * FROM Student;

UPDATE Student
SET Marks = 95
WHERE RollNo = 2;

DELETE FROM Student
WHERE RollNo = 3;
```

Expt. No. 8 Date

Page No. 8

Aim:
To study the use of WHERE, IN, BETWEEN, and LIKE clause in SQL.

Theory:
These clauses are used with the SELECT command to filter records from a table.

- WHERE → filters records based on ~~columns~~ conditions.
- IN → checks multiple values
- BETWEEN → selects values within a range
- LIKE → searches patterns in text.

Description:
- WHERE filters records with marks greater than 80
- IN selects students from Delhi & Mumbai
- BETWEEN selects marks b/w 70 and 90.
- LIKE 'A%' selects names starting with letter A.

Teacher's Signature

```
CREATE TABLE Student (
    RollNO INT,
    Name VARCHAR(50),
    Marks INT,
    city VARCHAR(30)
);
Insert INTO student VALUES
(1, 'Ali', 85, 'Delhi'),
(2, 'Ahmed', 90, 'Mumbai'),
(3, 'Zaid', 78, 'Delhi'),
(4, 'Umar', 67, 'Pune'),
(5, 'Bilal', 95, 'Mumbai');

SELECT * FROM Student
WHERE Marks > 80;

Select * from student
Where city IN ('Delhi', 'Mumbai');

SELECT * FROM Student
WHERE Marks BETWEEN 70 AND 90;

SELECT * FROM student
WHERE Name LIKE 'A%';
```

Date

Expt. No. 9 Page No. 9

Aim: To write an SQL query that returns the Project number and project name for projects having budget greater than 100000.

Theory: The SELECT command is used to retrieve data from a table. The WHERE clause is used to filter records according to a specified condition.

e.g. the Practical, records are selected from the PROJECT table where the budget is greater than 100000.

Teacher's Signature

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```
CREATE TABLE PROJECT (
    Pno INT,
    Pname VARCHAR(50),
    budget INT
    dno VARCHAR(10)
);
INSERT INTO PROJECT VALUES
('101', 'Banking System', 150000, 'D1'),
('102', 'Library System', 80000, 'D2'),
('103', 'Hospital System', 200000, 'D3');

SELECT Pno, Pname
FROM PROJECT
WHERE budget > 100000;
```

Expt. No 10 Date

Page No 10

Aim: To Study the use of AND, OR, and NOT operators in SQL queries.

Theory: Logical operators are used in SQL to combine multiple conditions.

- And → Returns Records ~~are used~~ when all conditions are true
- OR → Returns Records when any condition is true
- NOT → Returns Records, when the condition is false.

CREATE Table Employee (

Reference Page 3 Expt. No. 3

Select * from Employee
Where Department = 'IT' And Salary
> 30000;

Select * from Employee
Where Department = "HR" OR
Salary > 35000 ;

Select * from Employee
Where NOT Department = 'finance';

Date

Expt. No. 11

Page No.

Aim: To perform aggregate functions in SQL using COUNT, SUM, AVG, MAX, MIN, GROUP BY, and Having clauses.

Theory: Aggregate functions are used to perform calculations on multiple rows of a table and return a single value.

- count() → Counts rows
- Sum() → Adds values
- Avg() → Finds average
- Max() → Finds maximum value
- Min() → Finds minimum value
- Group by → Groups rows with same values
- Having → Applies condition on groups

Teacher's Signature

Table Used

Emp (eno, ename, title, salary, dno)

1. Count function

```
SELECT COUNT(*) AS Total_Employees
FROM Emp;
```

2. Sum function

```
SELECT SUM(salary) AS Total_Salary
FROM Emp;
```

3. AVG function

```
SELECT AVG(Salary) As Average_Salary
FROM Emp;
```

4. Max

```
SELECT Max(Salary) As Highest_Salary
FROM Emp;
```

5. Min

```
Select Min(Salary) As lowest_salary
from Emp;
```

6. GROUP By

```
SELECT dno, Count(*) AS Total_
Employees FROM Emp
GROUP BY dno;
```

7. Having

```
SELECT dno, Avg(Salary) As
Avg_Salary FROM Emp
GROUP BY dno
Having Avg(Salary) > 3000;
```

Date

Expt. No. 12 Page No.

Aim: To Study and Perform INNER JOIN and LEFT JOIN operations in SQL.

Theory: A JOIN in SQL is used to combine records from two or more tables using a common field. Joins are very important in relational database where data is usually stored in different tables.

INNER JOIN: Returns only those records where matching values are present in both tables. Non-matching records are not displayed.

LEFT JOIN: Returns all records from the left table and the matching records from the right table. If no matching record exists in the right table, NULL values are shown.

Teacher's Signature

```
CREATE Table Dept (
  dno VARCHAR (10) PRIMARY key ,
  dname VARCHAR (30)
);
CREATE TABLE EMP (
   eno INT PRIMARY KEY,
   ename VARCHAR (30),
   dno VARCHAR (10)
);
Insert Into Emp values
(1, 'Ali' , 'D1' ),
(2, 'Sara', 'D2'),
(3, 'John', NULL );

Select Emp.eno, Emp.ename, Dept.dname
From Emp INNER JOHN Dept
ON Emp.dno = Dept.dno ;

SELECT Emp.eno, Emp.ename, Dept.dname
FROM Emp
LEFT JOIN Dept
ON Emp.dno = Dept.dno ;
```

Date

Expt. No. 13 Page No.

Aim: To study and perform RIGHT JOIN and CROSS JOIN operations in SQL.

Theory: Joins are used to combine data from multiple tables based on a common field.

RIGHT JOIN: Displays all records from the right table, and matching records from the left table. If no match is found in the left table, NULL values are displayed.

CROSS JOIN: Displays the Cartesian Product of two tables, it combines every row of the first table with every row of the second table.

These Joins are useful for retrieving related data from tables.

Teacher's Signature

```
CREATE TABLE DEPT (
dno VARCHAR(10),
dname VARCHAR(20)
);
CREATE TABLE EMP(
eno INT,
ename VARCHAR(20),
dno VARCHAR(10)
);
INSERT INTO DEPT VALUES
(1,'Ali','D1'),
(2,'Sara','D2');
-- Right Join
SELECT ename, dname FROM EMP
RIGHT JOIN DEPT ON EMP.dno = DEPT.dno;

-- CROSS Join

SELECT ename, dname FROM EMP
CROSS Join DEPT;
```

Date

Expt. No. 14

Page No.

Aim: To display the details of employees whose working hours are less than 10 and responsibility is Manager.

Theory: The WHERE clause in SQL is used to apply conditions on records. Multiple conditions can be used with logical operators like AND.

- AND operator returns records only when all conditions are true.
- This query helps in filtering specific records from a table.

Teacher's Signature

```
CREATE TABLE WORKSON (
Eno INT,
Pno INT,
Resp VARCHAR(20),
hours INT
);

INSERT INTO WORKSON Values
(1, 101, 'Manager', 8),
(2, 102, 'Engineer', 12),
(3, 103, 'Manager', 6);

SELECT * FROM WORKSON
WHERE hours < 10 AND
Resp = 'Manager';
```

Date

Expt. No. 15 Page No.

Aim: To display employees whose title is FE or SA and salary is greater than 35000.

Theory: The WHERE clause is used to filter records based on conditions.
- OR operator is used when any one condition can be true.
- AND operator is used when all conditions must be true

This query helps in retrieving specific employee records based on title and salary.

Teacher's Signature

```
CREATE TABLE EMP (
  eno INT,
  ename VARCHAR(10),
  title VARCHAR(10),
  Solary INT
);
INSERT INTO EMP VALUES
(1, 'Ali', 'EE', 40000),
(2, 'Sara', 'SA', 45000),
(3, 'John', 'Manager', 30000);

SELECT * FROM EMP
WHERE (title = 'EE' OR title
= 'SA') AND Solary > 35000;
```

Date

Expt. No. 16 Page No.

Aim:

To display employee of department D1 ordered by decreasing salary.

Theory:

The WHERE clause is used to select specific records from a table. The order BY clause is used to arrange records in ascending or descending order.

- DESC is used for descending order.
- This query helps in sorting employee records based on salary.

```
CREATE TABLE EMP(
eno INT,
ename VARCHAR(20),
Salary INT,
dno VARCHAR(20)
);

INSERT INTO EMP VALUES
(1,'Ali',50000,'D1')
(2,'Sara',40000,'D2')
(3,'John',30000,'D1');

SELECT * FORM EMP WHERE
dno='D1' ORDER BY Salary DESC;
```

Date

Expt. No. 17

Page No.

Aim:
To display department records ordered by ascending department name.

Theory:
The order By clause in SQL is used to sort records in ascending or descending order.
• ASC is used for ascending order.
• By default, records are arranged in ascending order.

This query helps in organizing department data alphabetically.

```
CREATE TABLE DEPT (
dno VARCHAR (10),
dname VARCHAR (20).
);

INSERT INTO DEPT VALUES
('D1', 'HR'),
('D2', 'Sales'),
('D3', 'Consulting');

SELECT * FROM DEPT ORDER BY
dname ASC;
```

Expt. No. 18 Date

Page No.

Aim:

To display Project name, hours worked, and Project number where working are greater than 10.

Theory:

The WHERE clause is used to filter records based on a condition.

- It helps in displaying only required records from a table.
- Conditions can be applied using comparison operators such as >, <, = etc

This query retrieves Project details where working hours are greater than 10.

```
CREATE TABLE WORKSON (
Pno INT,
Pname VARCHAR(20),
hours INT
);

INSERT INTO WORKSON VALUES
(101, 'Website', 12),
(102, 'APP', 8),
(103, 'Database', 15);

SELECT Pname, hours, Pno
FROM WORKSON
WHERE hours > 10;
```